

Orbit Systems

Products, Geometric Families, and Pratt Layers

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Abstract

We use the term *orbit system* for a minimal structure that separates four operations: incidence stratification, primitive extraction, symmetry quotienting, and asymptotic counting. The present version broadens the type space from the divisor poset to an arbitrary locally finite poset and allows vector-valued gauges. This is essential for two new constructions developed here.

First, Grassmann orbit systems are built from surjections $(\mathbb{Z}/n\mathbb{Z})^k \rightarrow (\mathbb{Z}/n\mathbb{Z})^r$ modulo GL_r . Their atoms are free direct summands, and their cumulative atom count has a polynomial prime-number law of degree $r(k-r)+1$. The Jordan system becomes the case $r=1$. Second, Pratt orbit systems use the coordinatewise Pratt order and its Moebius inversion rather than ordinary divisibility. The primitive Pratt layers have masses $C_k(d)$, while quotienting by coordinatewise units produces the multiplicative atom count

$$A_k^P(d) = \prod_{p^e \parallel d} ((e+1)^k - 1).$$

Its Dirichlet series has the form $\zeta(s)^{2^k-1} H_k(s)$, leading to a polylogarithmic-linear law

$$\sum_{d \leq X} A_k^P(d) \sim C_k^P X (\log X)^{2^k-2}.$$

We also connect the orbit formalism to ranked recursive factorization systems (RRFS). A multiplicative realization turns the RRFS telescoping reconstruction into a linear logarithmic functional on recursive coordinates; the coordinatewise Pratt order is locally finite, and Moebius inversion then produces an RRFS-primitive logarithmic transform. For the full polynomial RRFS over $\mathbb{Q}$, we take Mahler measure as the multiplicative norm. At fixed degree d , the classical Chern–Vaaler theory gives a polynomial Mahler prime-number law of order T^d . We then define a genuinely Pratt-geometric Chebyshev function by Moebius-inverting $\log \mathcal{M}$ on the polynomial Pratt poset. SymPy experiments in degrees 1–4 suggest a new fixed-degree law of order T^d for $d \geq 2$, with degree 1 exceptional of order $T \log T$.

We also define the Cartesian product of orbit systems. Primitive layers, orbit spaces, Moebius functions, and atom counts tensor exactly, giving a systematic method for producing new growth laws from old ones. Existing examples—finite fields, necklaces, periodic dynamics, Jordan systems, cyclotomic roots, and the Pratt-measured integer model—are retained to show how the expanded definition specializes. The goal is not to claim novelty for the classical ingredients, but to make the adjustable mechanisms explicit and to exhibit families whose orbit laws are generated by the formalism itself.

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1 Purpose and scope

Many objects called primes, irreducibles, primitive cycles, or primitive orbit classes are obtained by a common sequence:

$$\boxed{\text{total layer} \longrightarrow \text{primitive type} \longrightarrow \text{symmetry quotient} \longrightarrow \text{orbit atoms} \longrightarrow \text{growth law}.}$$

The point of an orbit system is to record these steps without identifying them with ordinary multiplicative primality. The type space need not be the divisor lattice. It may be a Pratt poset, a product poset, a subspace lattice, or another locally finite incidence geometry.

Three structures must be kept distinct:

$$\text{incidence/Moebius stratification} \neq \text{symmetry quotient} \neq \text{asymptotic gauge}.$$

A fourth structure, recursive factorization, may be added later but is not part of the definition.

We use *prime-number law* as an umbrella term for a leading asymptotic of primitive orbit atoms. It is always qualified by its scale: *norm-logarithmic*, *exponential layer*, *polynomial cumulative*, *polylogarithmic-linear*, or another explicitly stated form. This terminology records a growth law; it does not assert that all examples are arithmetically equivalent to the ordinary primes.

A central theme of the present paper is constructive. Once an orbit system is given, products and geometric replacements of the primitive orbit space generate further systems. Thus the formalism is intended not only to classify familiar examples but also to produce new test families.

1.1 What is and is not being claimed

The individual formulas used below are classical. In particular:

- divisor-poset Moebius inversion is classical;
- the necklace polynomial counts both primitive necklaces and monic irreducible polynomials over finite fields;
- fixed-point counts and primitive periodic orbits are classically related by Moebius inversion, and prime orbit theorems form an established part of dynamics;
- Burnside/ghost coordinates, necklace algebras, and Witt vectors have a substantial established theory;
- Jordan totients, the Dedekind psi function, and their Dirichlet series are classical arithmetic objects.

The proposed contribution is the four-axis packaging and the resulting comparison principle: the same primitive-quotient mechanism may support different asymptotic regimes because the total-layer growth, orbit weight, and gauge are independent inputs. We do not claim priority for the phrase “orbit system”; we use it for the definition below.

The Grassmann and Pratt families below are constructions inside this language. Their ingredients (Grassmannians over finite rings, Pratt coordinates, incidence Moebius inversion, and Selberg–Delange type asymptotics) have independent antecedents. What is asserted here is the resulting orbit-system construction and the deductions proved from the stated formulas, not a priority claim that none of the component counting functions has appeared elsewhere.

2 The four axioms

Let P be a locally finite poset, written $s \preceq t$. We permit a gauge with several coordinates because Cartesian products naturally carry independent size variables.

Definition 2.1 (Orbit system). An *orbit system* is data

$$\mathcal{X} = (P, \{(X_t, \mu_t)\}_{t \in P}, \{(E_t, G_t)\}_{t \in P}, \nu)$$

satisfying the following four axioms.

OS1. Measured layers. For every $t \in P$, X_t is a finite set equipped with a nonnegative additive measure μ_t on subsets.

OS2. Primitive stratification. For every t there is a disjoint measured decomposition

$$X_t = \bigsqcup_{s \preceq t} E_s^{(t)},$$

where $E_s^{(t)}$ is canonically identified, as a measured set, with a primitive model E_s independent of t .

OS3. Orbit atomization. A finite group G_s acts measure-preservingly on E_s . The type- s atom space is

$$\mathcal{A}_s := E_s / G_s.$$

For an orbit $a \in \mathcal{A}_s$, its weight is $w(a) := \mu_s(a)$.

OS4. Proper gauge. A gauge

$$\nu : \bigsqcup_{s \in P} \mathcal{A}_s \longrightarrow \mathbb{R}_{\geq 0}^m$$

is specified for some finite $m \geq 1$, with coordinatewise order, such that every bounded box contains only finitely many atoms. Scalar gauges are the case $m = 1$.

Define

$$F(t) := \mu_t(X_t), \quad P_0(t) := \mu(E_t) = \sum_{a \in \mathcal{A}_t} w(a), \quad a(t) := \#\mathcal{A}_t.$$

Axiom OS2 gives the master identity

$$F(t) = \sum_{s \preceq t} P_0(s). \tag{1}$$

Theorem 2.2 (Primitive-layer inversion). *Let μ_P be the Moebius function of P . Then*

$$P_0(t) = \sum_{s \preceq t} \mu_P(s, t) F(s). \tag{2}$$

If every orbit in E_t has the same weight $w(t)$, then

$$a(t) = \frac{1}{w(t)} \sum_{s \preceq t} \mu_P(s, t) F(s).$$

Proof. Equation (1) is the zeta transform in the incidence algebra of P . Applying its inverse gives (2); the homogeneous-orbit formula is $P_0(t) = w(t)a(t)$. $\square$

For the divisor poset $(\mathbb{N}, |)$, (2) becomes

$$w(n)a(n) = \sum_{d|n} \mu(d) F(n/d) \tag{3}$$

when the primitive orbit weight depends only on the type.

Definition 2.3 (Prime-number law). A *prime-number law* is a proved leading asymptotic for a layer count $a(t)$, a cumulative atom count, or a cumulative primitive mass, with the gauge explicitly stated. For a scalar type gauge one may write

$$A(Y) = \sum_{\nu(t) \leq Y} a(t), \quad \Psi(Y) = \sum_{\nu(t) \leq Y} P_0(t).$$

For a vector gauge one uses rectangular counts $A(Y_1, \dots, Y_m)$.

3 Ranked recursive factorization systems and induced orbit systems

We now recall the recursive structure that motivates the Pratt gauges used later. This is additional structure: an orbit system does not require an RRFS, but an RRFS with a positive multiplicative realization can canonically supply a locally finite coordinate order and a logarithmic measure.

Definition 3.1 (Ranked recursive factorization system). A *ranked recursive factorization system* (RRFS) is a quadruple

$$\mathfrak{R} = (H, \mathcal{B}, D, \rho),$$

where H is a reduced factorial commutative monoid with atom set $\mathcal{B}$, $D : \mathcal{B} \rightarrow H$ is a predecessor map, and $\rho : \mathcal{B} \rightarrow \mathbb{N}_0$ is a rank such that

$$\beta \mid D\alpha \implies \rho(\beta) < \rho(\alpha).$$

Writing

$$D\alpha = \prod_{\beta \in \mathcal{B}} \beta^{B_{\beta, \alpha}},$$

rank descent makes the recursive tree of every atom finite. For $h \in H$, let $M_\beta(h)$ denote the total number of β -labelled vertices in the fully unfolded recursive forest of h .

If $v(h)$ is the ordinary root multiplicity vector, then on every finite rank-closed truncation

$$M(h) = (I - B)^{-1}v(h), \quad v(h) = (I - B)M(h).$$

In particular, M is injective. In the quotient group $q(H)$ one has the telescoping reconstruction

$$h = \prod_{\alpha \in \mathcal{B}} \left(\frac{\alpha}{D\alpha} \right)^{M_\alpha(h)}. \quad (4)$$

These are the basic RRFS identities used below; they are recalled here only to make the present paper self-contained at the point where RRFS and orbit systems meet.

3.1 The RRFS Pratt order and logarithmic functional

Assume that a multiplicative realization

$$N : q(H) \longrightarrow \mathbb{R}_{>0}$$

is given. Put

$$\varepsilon_\alpha := \log \frac{N(\alpha)}{N(D\alpha)} \quad (5)$$

and define the RRFS Pratt order by

$$a \preceq_{\mathfrak{R}} b \iff M_\alpha(a) \leq M_\alpha(b) \text{ for every } \alpha \in \mathcal{B}. \quad (6)$$

Because $M(b)$ has finite support, the interval below b contains only finitely many possible coordinate vectors; injectivity of M therefore shows that $(H, \preceq_{\mathfrak{R}})$ is locally finite.

Applying N to (4) and taking logarithms gives the basic bridge.

Theorem 3.2 (RRFS logarithmic functional). *For every $h \in H$,*

$$\log N(h) = \sum_{\alpha \in \mathcal{B}} M_\alpha(h) \varepsilon_\alpha. \quad (7)$$

If every $\varepsilon_\alpha \geq 0$, then $a \preceq_{\mathfrak{R}} b$ implies $N(a) \leq N(b)$.

Proof. The first assertion is the logarithm of (4). The monotonicity assertion follows because (7) is then a nonnegative linear functional of the coordinate vector. $\square$

Let $\mu_{\mathfrak{R}}$ be the Moebius function of the locally finite Pratt poset. Define

$$\Lambda_{\mathfrak{R}}(h) := \sum_{a \preceq_{\mathfrak{R}} h} \mu_{\mathfrak{R}}(a, h) \log N(a). \quad (8)$$

Then incidence inversion gives the exact identity

$$\log N(h) = \sum_{a \preceq_{\mathfrak{R}} h} \Lambda_{\mathfrak{R}}(a). \quad (9)$$

Thus an RRFS provides, from its own recursive coordinates, both a locally finite incidence geometry and a canonical logarithmic total measure. If $\Lambda_{\mathfrak{R}}(h) \geq 0$ for all h , then by taking a weighted singleton primitive model of mass $\Lambda_{\mathfrak{R}}(h)$ at each type, Theorem 3.5 turns this data into an orbit system with trivial orbit groups. More generally, any intrinsic finite group realization of these primitive masses may replace the singleton model.

Remark 3.3 (Moebius positivity). Nonnegativity of the local coefficients ε_{α} does not by itself imply $\Lambda_{\mathfrak{R}} \geq 0$. Hence *Pratt-Moebius positivity*

$$\Lambda_{\mathfrak{R}}(h) \geq 0 \quad (h \in H)$$

is a genuine additional property of an RRFS realization. The construction above therefore separates two questions: recursive reconstruction always gives the logarithmic functional, while positivity decides whether its Pratt-poset primitive transform is itself a measured orbit system.

3.2 Polynomial RRFS: Mahler geometry and Pratt geometry

We now use the full polynomial RRFS rather than a distinguished recursively indexed subfamily. Let H_{poly} be the multiplicative monoid of monic polynomials in $\mathbb{Q}[x]$, with monic irreducible atoms ϕ . The predecessor is

$$D\phi = \phi - \phi(0), \quad Dx = 1.$$

Factoring $D\phi$ into monic irreducibles gives the children of ϕ . The recursive coordinates $M_{\psi}(f)$ satisfy the RRFS reconstruction

$$f = \prod_{\psi} \left(\frac{\psi}{D\psi} \right)^{M_{\psi}(f)} \quad (10)$$

in the quotient group. The terminal coordinate is distinguished by

$$M_x(f) = \deg f. \quad (11)$$

Mahler geometry. For a monic polynomial $f(x) = \prod_{j=1}^d (x - \alpha_j)$ define

$$\mathcal{M}(f) := \prod_{j=1}^d \max\{1, |\alpha_j|\}.$$

Mahler measure is multiplicative. Hence applying $\mathcal{M}$ to (10) gives the exact Pratt-linear identity

$$\log \mathcal{M}(f) = \sum_{\psi} M_{\psi}(f) \log \frac{\mathcal{M}(\psi)}{\mathcal{M}(D\psi)}. \quad (12)$$

Unlike the numerical coefficients $\log(p/(p-1))$, the polynomial local coefficients in (12) need not be nonnegative. Thus Mahler measure gives a canonical multiplicative norm and an exact signed Pratt functional, but not automatically a positive node measure.

Mahler measure is not proper on all irreducibles when the degree is allowed to vary, because infinitely many cyclotomic irreducibles have Mahler measure 1. The natural prime-number law is therefore fixed-degree. For $d \geq 1$ put

$$\pi_{\mathcal{M},d}(T) := \#\{\phi \in \mathbb{Z}[x] : \phi \text{ monic irreducible, } \deg \phi = d, \mathcal{M}(\phi) \leq T\}. \quad (13)$$

The lattice-point results of Chern and Vaaler, together with the lower-order contribution of reducible monic polynomials, give an explicit constant $C_d > 0$ such that

$$\pi_{\mathcal{M},d}(T) \sim C_d T^d \quad (T \rightarrow \infty, d \text{ fixed}). \quad (14)$$

We call (14) the *fixed-degree Mahler prime-number law*. The terminology refers only to atom growth in the present orbit-system sense; the counting theorem itself is classical [13, 14].

Pratt geometry. The same polynomial universe carries the coordinatewise Pratt order

$$f \preceq_P g \iff M_{\psi}(f) \leq M_{\psi}(g) \text{ for every irreducible } \psi. \quad (15)$$

Let μ_P be its Moebius function. Define the Pratt-primitive Mahler transform

$$\Lambda_{P,\mathcal{M}}(f) := \sum_{g \preceq_P f} \mu_P(g, f) \log \mathcal{M}(g). \quad (16)$$

Then incidence inversion gives

$$\log \mathcal{M}(f) = \sum_{g \preceq_P f} \Lambda_{P,\mathcal{M}}(g). \quad (17)$$

Thus Mahler geometry supplies a proper fixed-degree cutoff, while Pratt geometry decides which part of that logarithmic mass is primitive with respect to recursive factorization.

For fixed degree d define the Pratt–Mahler Chebyshev function

$$\Psi_{P,d}(T) := \sum_{\substack{f \in \mathbb{Z}[x] \text{ monic, } \deg f = d \\ \mathcal{M}(f) \leq T}} \Lambda_{P,\mathcal{M}}(f). \quad (18)$$

A *Pratt-geometric prime-number theorem* is an asymptotic law for (18). It is not a formal consequence of (14): the two functions probe different structures on the same polynomials.

3.2.1 SymPy experiment and an empirical conjecture

We performed a finite exact/computational experiment with SymPy 1.14.0. For every tested monic integer polynomial f of degree d and Mahler measure at most T , the program factored f over $\mathbb{Q}$, recursively factored $D\phi = \phi - \phi(0)$, constructed the recursive coordinate vector and transition matrix, enumerated the finite interval $0 \preceq_P g \preceq_P f$ via the RRFS cone condition $(I - B)M(g) \geq 0$, and performed Moebius inversion of $\log \mathcal{M}$ on that interval.

Candidate polynomials were exhaustively enumerated using the standard coefficient consequence of Mahler's bound

$$|a_{d-j}| \leq \binom{d}{j} \mathcal{M}(f),$$

and then filtered by direct numerical evaluation of the root-product defining $\mathcal{M}(f)$. Thus each displayed cutoff is an exhaustive finite computation, not a random sample. It is evidence only, not an asymptotic proof.

d	T	tested monic f	$\Psi_{P,d}(T)$	$\Psi_{P,d}(T)/T^d$
1	10	21	25.603	2.560
1	50	101	296.956	5.939
1	100	201	727.479	7.275
2	4	75	24.080	1.505
2	8	275	105.844	1.654
2	12	603	246.020	1.708
2	15	933	394.690	1.754
2	20	1643	713.643	1.784
3	2	89	10.961	1.370
3	3	267	41.574	1.540
3	4	593	88.656	1.385
3	5	1123	168.373	1.347
3	6	1897	267.578	1.239
3	7	2967	424.283	1.237
4	2	307	19.712	1.232

Degree 1 can be analyzed exactly. Every monic linear polynomial is $x - a$, with $D(x - a) = x$ and

$$\Lambda_{P,\mathcal{M}}(x - a) = \log \max(1, |a|).$$

Consequently

$$\Psi_{P,1}(T) = 2 \sum_{2 \leq a \leq T} \log a = 2 \log([T]!) \sim 2T \log T. \quad (19)$$

Thus degree 1 is genuinely logarithmically enhanced.

For $d = 2$, the ratio $\Psi_{P,2}(T)/T^2$ rises slowly from 1.50 at $T = 4$ to 1.78 at $T = 20$, whereas division by $T^2 \log T$ decreases steadily. For $d = 3$, the ratio to T^3 is roughly stable near 1.2–1.4 over the largest tested range. Degree 4 at $T = 2$ is compatible with the same power law but is too small a range to be probative. Individual primitive values are signed: positive, zero, and negative values all occur from degree 2 onward. The apparent power law is therefore a cancellation phenomenon rather than a positivity phenomenon.

Conjecture 3.4 (Pratt–Mahler PNT, empirical form). *For every fixed degree $d \geq 2$ there exists a real constant K_d such that*

$$\Psi_{P,d}(T) \sim K_d T^d \quad (T \rightarrow \infty). \quad (20)$$

For $d = 1$, instead, (19) gives $\Psi_{P,1}(T) \sim 2T \log T$.

The guess (20) is intentionally conservative. It says that Pratt Moebius inversion appears to preserve the Mahler polynomial exponent d for $d \geq 2$ while changing the constant and creating substantial signed cancellation. A stronger theorem would identify K_d , establish whether it is positive for every $d \geq 2$, and give an error term. A vector-valued refinement would ask for the distribution of normalized Pratt profiles inside Mahler balls.

3.3 The free layer construction

The axioms do not require that the total layers exist before the primitive models are known.

Proposition 3.5 (Free orbit-system construction). *Let P be locally finite. Suppose that for each $s \in P$ one is given a finite measured G_s -set E_s . Define*

$$X_t := \bigsqcup_{s \preceq t} E_s$$

with the disjoint-union measure. Then OS1–OS3 hold automatically. Any proper gauge on the orbit set completes an orbit system.

This elementary observation is useful for Grassmann and Pratt examples: the primitive geometry is intrinsic, while the total layer may be regarded as its incidence closure.

3.4 Cartesian products

Let $\mathcal{X}_1$ and $\mathcal{X}_2$ be orbit systems over P_1 and P_2 . Put the product order on $P_1 \times P_2$.

Definition 3.6 (Cartesian product). The Cartesian product $\mathcal{X}_1 \times \mathcal{X}_2$ has

$$X_{(s,t)} := X_s^{(1)} \times X_t^{(2)}, \quad E_{(s,t)} := E_s^{(1)} \times E_t^{(2)},$$

product measure, group

$$G_{(s,t)} := G_s^{(1)} \times G_t^{(2)},$$

and concatenated gauge

$$\nu(a_1, a_2) := (\nu_1(a_1), \nu_2(a_2)).$$

Theorem 3.7 (Product theorem). *The Cartesian product is an orbit system. Moreover,*

$$\begin{aligned} F_{12}(s, t) &= F_1(s)F_2(t), & P_{0,12}(s, t) &= P_{0,1}(s)P_{0,2}(t), \\ \mathcal{A}_{12}(s, t) &\cong \mathcal{A}_1(s) \times \mathcal{A}_2(t), & a_{12}(s, t) &= a_1(s)a_2(t). \end{aligned}$$

The Moebius function factors:

$$\mu_{P_1 \times P_2}((r, u), (s, t)) = \mu_{P_1}(r, s)\mu_{P_2}(u, t).$$

For rectangular cumulative counts,

$$A_{12}(Y_1, Y_2) = A_1(Y_1)A_2(Y_2).$$

Proof. The product of the two disjoint primitive decompositions is the disjoint decomposition indexed by pairs. Product groups have product orbits because

$$(E_s^{(1)} \times E_t^{(2)}) / (G_s^{(1)} \times G_t^{(2)}) \cong (E_s^{(1)} / G_s^{(1)}) \times (E_t^{(2)} / G_t^{(2)}).$$

The measure and counting identities follow. The factorization of the Moebius function is the standard tensor-product identity for incidence algebras. $\square$

Corollary 3.8 (Growth multiplication). *If $A_i(Y) \sim C_i Y^{\alpha_i} (\log Y)^{\beta_i}$ and the product is counted in a common box $\nu_1, \nu_2 \leq Y$, then*

$$A_{12}(Y, Y) \sim C_1 C_2 Y^{\alpha_1 + \alpha_2} (\log Y)^{\beta_1 + \beta_2}.$$

Thus Cartesian products add polynomial exponents and logarithmic exponents.

4 The growth mechanism

The axioms do not imply a universal asymptotic. They only reduce the counting problem to two successive operations:

$$F \xrightarrow{\text{Moebius inversion}} P \xrightarrow{\text{divide by orbit weight}} a.$$

The gauge then determines which cumulative count is natural.

4.1 Exponential total layers

The following estimate is a standard consequence of divisor Moebius inversion. It is included because it explains simultaneously the finite-field, necklace, and elementary dynamical examples.

Proposition 4.1 (Standard exponential inversion estimate). *Suppose $P = (\mathbb{N}, |)$, every primitive orbit in type n has weight $w(n) = n$, and*

$$F(n) = c\lambda^n + O(\theta^n), \quad c > 0, \quad \lambda > 1, \quad 0 \leq \theta < \lambda.$$

Then

$$a(n) = \frac{c\lambda^n}{n} + O\left(\frac{\theta^n + \tau(n)\lambda^{n/2}}{n}\right),$$

and in particular

$$a(n) \sim \frac{c\lambda^n}{n}.$$

Proof. By (3),

$$a(n) = \frac{1}{n} \sum_{d|n} \mu(d) F(n/d).$$

The term $d = 1$ is $c\lambda^n/n + O(\theta^n/n)$. Every divisor $d > 1$ satisfies $n/d \leq n/2$, and $F(m) = O(\lambda^m)$, giving the displayed error. $\square$

The occurrence of λ^n/n is therefore structural whenever the total layer is exponential and the primitive symmetry has orbit size n .

4.2 Polynomial total layers

Polynomial layers behave differently. Proper divisors are not exponentially suppressed, so a pointwise main term need not emerge from the $d = 1$ term. Jordan systems illustrate this: the natural law is cumulative and its constant is an Euler product.

4.3 Measured logarithmic layers

The ordinary integers present a third regime. Their numerical Pratt coordinates carry the positive linear functional

$$L_P(M(n)) = \sum_q m_q(n) \log \frac{q}{q-1} = \log n.$$

Using this intrinsic Pratt mass as the total-layer measure, ordinary divisor-poset Moebius inversion gives the sparse primitive transform Λ , supported on prime powers. The prime number theorem then enters as an additional analytic theorem about the gauge, not as a formal consequence of OS1–OS4.

5 Finite fields: Frobenius quotients

Fix a prime power q . Let $P = (\mathbb{N}, |)$ and

$$X_n = \mathbb{F}_{q^n}$$

with counting measure. For $d \mid n$, let $E_d^{(n)}$ be the elements of exact degree d over $\mathbb{F}_q$. The primitive model is

$$E_d = \{\alpha \in \overline{\mathbb{F}_q} : [\mathbb{F}_q(\alpha) : \mathbb{F}_q] = d\}.$$

The cyclic group generated by Frobenius

$$\text{Fr}_q(\alpha) = \alpha^q$$

acts freely on E_d , and every orbit has cardinality d . The orbits are in bijection with monic irreducible polynomials of degree d over $\mathbb{F}_q$. Write their number as $I_q(d)$. Thus

$$q^n = \sum_{d \mid n} d I_q(d), \tag{21}$$

and

$$I_q(n) = \frac{1}{n} \sum_{d \mid n} \mu(d) q^{n/d}. \tag{22}$$

Theorem 5.1 (Exponential finite-field law). *For fixed q ,*

$$I_q(n) = \frac{q^n}{n} + O\left(\frac{\tau(n)q^{n/2}}{n}\right), \quad I_q(n) \sim \frac{q^n}{n}.$$

Proof. Apply Theorem 4.1 with $F(n) = q^n$, $c = 1$, $\lambda = q$. □

This formula is classical. In the present framework its role is explanatory: exponential total layer q^n , divisor stratification, and a free cyclic symmetry of size n force the scale q^n/n .

6 Necklaces and Lyndon words

Let B be an alphabet of size $q \geq 2$ and $X_n = B^n$. A word of length n has a unique least period $d \mid n$ and is obtained by repeating a primitive word of length d exactly n/d times. Let E_d be the primitive (aperiodic) words of length d .

The cyclic group C_d acts freely on E_d by rotation. Its orbit atoms are primitive necklaces. Let $M_q(d)$ denote their number. Then

$$q^n = \sum_{d \mid n} d M_q(d), \quad M_q(n) = \frac{1}{n} \sum_{d \mid n} \mu(d) q^{n/d}. \quad (23)$$

Consequently

$$M_q(n) \sim \frac{q^n}{n}.$$

If the alphabet is ordered, each primitive necklace has a unique lexicographically least rotation, a Lyndon word, so primitive necklaces and Lyndon words have the same count.

Comparing (22) and (23),

$$M_q(n) = I_q(n)$$

numerically for every q, n . This identity is classical and is encoded by the necklace polynomial. No canonical identification between a word and an irreducible polynomial is asserted here. What the orbit-system viewpoint explains is why two different geometries have the same count: they have the same total function $F(n) = q^n$ and the same primitive orbit weight $w(n) = n$.

7 Periodic dynamics and graph cycles

7.1 Finite fixed-point systems

Let $T : Y \rightarrow Y$ be a map such that $\# \text{Fix}(T^n) < \infty$ for every n . Set

$$X_n = \text{Fix}(T^n), \quad F_T(n) = \# \text{Fix}(T^n).$$

The primitive type- d layer consists of points of least period d . The group C_d acts freely by iteration. If $P_T(d)$ is the number of primitive periodic orbits of least period d , then

$$F_T(n) = \sum_{d \mid n} d P_T(d), \quad P_T(n) = \frac{1}{n} \sum_{d \mid n} \mu(d) F_T(n/d). \quad (24)$$

This is the standard periodic-orbit transform underlying the Artin–Mazur zeta function

$$\zeta_T(z) = \exp \left(\sum_{n \geq 1} \frac{F_T(n)}{n} z^n \right) = \prod_{\tau} (1 - z^{|\tau|})^{-1},$$

where τ runs over primitive periodic orbits whenever the formal product is defined.

If $F_T(n) = c\lambda^n + O(\theta^n)$ with $\theta < \lambda$, Theorem 4.1 gives

$$P_T(n) \sim \frac{c\lambda^n}{n}.$$

This elementary implication is not proposed as a new prime-orbit theorem. Prime orbit theorems in hyperbolic and symbolic dynamics form a classical and substantially deeper analytic theory; here the estimate serves only as a reusable component of the comparison.

Finite-field Frobenius and the full shift on q symbols are special cases with $F(n) = q^n$.

7.2 Primitive cycles in finite directed graphs

Let B be a primitive nonnegative integral matrix, viewed as the adjacency matrix of a finite directed graph or edge shift. The number of based closed walks of length n is

$$F_B(n) = \text{tr}(B^n).$$

Cyclic rotation quotients least-period closed walks into primitive cycles, so

$$P_B(n) = \frac{1}{n} \sum_{d|n} \mu(d) \text{tr}(B^{n/d}).$$

If $\lambda > 1$ is the Perron–Frobenius eigenvalue of B , primitivity implies that every other eigenvalue has modulus $< \lambda$, hence

$$\text{tr}(B^n) = \lambda^n + O(\theta^n)$$

for some $\theta < \lambda$. Therefore

$$P_B(n) \sim \frac{\lambda^n}{n}.$$

This is the exponential layer law for primitive graph cycles.

8 Jordan systems: a polynomial law

Fix $k \geq 1$ and let

$$X_n = (\mathbb{Z}/n\mathbb{Z})^k$$

with counting measure. Every element has exact additive order $d \mid n$. The primitive type- d model is

$$E_k(d) = \{(a_1, \dots, a_k) \in (\mathbb{Z}/d\mathbb{Z})^k : \gcd(a_1, \dots, a_k, d) = 1\}.$$

Its cardinality is Jordan’s totient

$$J_k(d) = d^k \prod_{p|d} (1 - p^{-k}),$$

so

$$n^k = \sum_{d|n} J_k(d). \tag{25}$$

The unit group $U_d = (\mathbb{Z}/d\mathbb{Z})^\times$ acts diagonally on $E_k(d)$. The action is free. Indeed, if $ua = a$, then $(u - 1)a_i \equiv 0 \pmod{d}$ for every i ; primitivity forces $u \equiv 1 \pmod{d}$. Thus each orbit has cardinality $\varphi(d)$, and the orbit-atom count is

$$A_k(d) := \frac{J_k(d)}{\varphi(d)}. \tag{26}$$

Hence

$$n^k = \sum_{d|n} \varphi(d) A_k(d). \quad (27)$$

For prime powers,

$$A_k(p^e) = p^{(e-1)(k-1)} (1 + p + \cdots + p^{k-1}). \quad (28)$$

In particular,

$$A_k(p) = 1 + p + \cdots + p^{k-1} = \#\mathbb{P}^{k-1}(\mathbb{F}_p).$$

For $k = 2$, $A_2(n)$ is the classical Dedekind psi function.

8.1 Dirichlet factorization

The function A_k is multiplicative. Its Dirichlet series is

$$\mathcal{F}_k(s) = \sum_{n \geq 1} \frac{A_k(n)}{n^s}.$$

Using (28),

$$\mathcal{F}_{k,p}(s) = 1 + \frac{(1 + p + \cdots + p^{k-1})p^{-s}}{1 - p^{k-1-s}} = \frac{1 + (1 + p + \cdots + p^{k-2})p^{-s}}{1 - p^{k-1-s}}.$$

Therefore

$$\mathcal{F}_k(s) = \zeta(s - k + 1) G_k(s), \quad G_k(s) = \prod_p \left(1 + \frac{1 + p + \cdots + p^{k-2}}{p^s} \right), \quad (29)$$

with the numerator interpreted as 0 when $k = 1$. The Euler product $G_k(k)$ converges absolutely because each local perturbation is $O(p^{-2})$.

Theorem 8.1 (Polynomial Jordan law). *For fixed $k \geq 1$,*

$$\sum_{n \leq X} A_k(n) \sim C_k X^k,$$

where

$$C_k = \frac{1}{k} \prod_p \left(1 + \frac{1 + p + \cdots + p^{k-2}}{p^k} \right). \quad (30)$$

For $k = 1$, $C_1 = 1$. For $k = 2$,

$$C_2 = \frac{1}{2} \prod_p (1 + p^{-2}) = \frac{15}{2\pi^2}.$$

Proof. Write $G_k(s) = \sum_{d \geq 1} g_k(d) d^{-s}$. The coefficients are nonnegative, multiplicative, supported on squarefree integers, and satisfy

$$g_k(p) = 1 + p + \cdots + p^{k-2}.$$

Equation (29) gives

$$A_k = \text{id}_{k-1} * g_k, \quad \text{id}_{k-1}(m) = m^{k-1}.$$

Hence

$$\sum_{n \leq X} A_k(n) = \sum_{d \leq X} g_k(d) \sum_{m \leq X/d} m^{k-1}.$$

Since

$$\sum_{m \leq y} m^{k-1} = \frac{y^k}{k} + O_k(y^{k-1} + 1),$$

we obtain

$$\sum_{n \leq X} A_k(n) = \frac{X^k}{k} \sum_{d \leq X} \frac{g_k(d)}{d^k} + o(X^k).$$

To justify the error, use the convergence of $\sum g_k(d)d^{-k} = G_k(k)$ and split the error sums at a fixed M : the part $d \leq M$ is $o(X^k)$ as $X \rightarrow \infty$, while the tail is bounded by the convergent tail of $\sum g_k(d)d^{-k}$. Finally

$$\sum_{d \leq X} \frac{g_k(d)}{d^k} \longrightarrow G_k(k),$$

which gives (30). □

The function $A_k = J_k/\varphi$ is classical; the point here is its interpretation as a quotient of primitive vectors by scalar units and its placement beside the exponential orbit systems. Unlike the exponential cases, the natural leading statement is cumulative rather than pointwise.

9 Grassmann orbit systems

Fix integers $1 \leq r < k$ and put

$$D := r(k - r).$$

For $n \geq 1$ let $R_n = \mathbb{Z}/n\mathbb{Z}$ and define the primitive set

$$E_{k,r}(n) := \text{Surj}_{R_n}(R_n^k, R_n^r).$$

The group $G_n = \text{GL}_r(R_n)$ acts by postcomposition. The action is free: if $gT = T$ and T is surjective, then $g = 1$. Two surjections are in the same orbit exactly when they have the same kernel. Hence the atoms can be identified with rank- $(k - r)$ direct summands of R_n^k , equivalently with the finite-ring Grassmannian

$$\mathcal{A}_{k,r}(n) \cong \text{Gr}_{k-r}(R_n^k).$$

Using the free construction of theorem 3.5 over the divisor poset gives an orbit system. We use the type gauge $\nu(n) = n$.

9.1 Local atom count

Write the Gaussian binomial coefficient as

$$\begin{bmatrix} k \\ r \end{bmatrix}_p = \prod_{j=0}^{r-1} \frac{p^k - p^j}{p^r - p^j}.$$

It counts r -dimensional quotients, or equivalently $(k - r)$ -dimensional subspaces, of $\mathbb{F}_p^k$.

Proposition 9.1 (Grassmann count over prime powers). *For every prime p and $e \geq 1$,*

$$a_{k,r}(p^e) = p^{(e-1)D} \begin{bmatrix} k \\ r \end{bmatrix}_p.$$

Consequently $a_{k,r}$ is multiplicative and

$$a_{k,r}(n) = \prod_{p^e \parallel n} p^{(e-1)D} \begin{bmatrix} k \\ r \end{bmatrix}_p.$$

Proof. Reduction modulo p maps direct summands of $R_{p^e}^k$ onto subspaces of $\mathbb{F}_p^k$. Fix such a subspace and a complementary lift. Every lift of the chosen summand is the graph of a matrix whose entries lie in pR_{p^e} , modulo the corresponding change at the previous level. At each lift from p^j to p^{j+1} there are $p^{r(k-r)} = p^D$ choices. Starting with $\begin{bmatrix} k \\ r \end{bmatrix}_p$ subspaces modulo p gives the formula. The Chinese remainder theorem gives the product over distinct primes. $\square$

For $r = 1$ this recovers the Jordan/projective orbit count

$$a_{k,1}(p^e) = p^{(e-1)(k-1)}(1 + p + \cdots + p^{k-1}).$$

Thus the Jordan example is the projective edge of a Grassmann family.

9.2 Worked Grassmann examples

For $(k, r) = (3, 1)$ one has $D = 2$ and

$$a_{3,1}(p^e) = p^{2(e-1)}(p^2 + p + 1).$$

Hence there are 7 atoms over $\mathbb{F}_2$, 28 atoms over $\mathbb{Z}/4\mathbb{Z}$, and the cumulative law has degree 3.

For $(k, r) = (4, 2)$, $D = 4$ and

$$\begin{bmatrix} 4 \\ 2 \end{bmatrix}_p = p^4 + p^3 + 2p^2 + p + 1 = (p^2 + 1)(p^2 + p + 1).$$

Thus

$$a_{4,2}(2) = 35, \quad a_{4,2}(4) = 2^4 \cdot 35 = 560,$$

and the cumulative law has degree 5. These are genuinely different atom spaces from projective Jordan atoms: an atom is now a rank-2 direct summand, not a line.

9.3 Polynomial Grassmann PNT

Set

$$Z_{k,r}(s) = \sum_{n \geq 1} \frac{a_{k,r}(n)}{n^s}.$$

By theorem 9.1,

$$Z_{k,r,p}(s) = 1 + \begin{bmatrix} k \\ r \end{bmatrix}_p \frac{p^{-s}}{1 - p^{D-s}}.$$

Therefore

$$Z_{k,r}(s) = \zeta(s - D)H_{k,r}(s), \quad (31)$$

where

$$H_{k,r}(s) = \prod_p \left(1 + \left(\begin{bmatrix} k \\ r \end{bmatrix}_p - p^D \right) p^{-s} \right).$$

Since $\begin{bmatrix} k \\ r \end{bmatrix}_p = p^D + O_{k,r}(p^{D-1})$, the Euler product for $H_{k,r}$ converges absolutely in a neighborhood of $s = D + 1$.

Theorem 9.2 (Grassmann prime-number law). *For fixed k, r ,*

$$\sum_{n \leq X} a_{k,r}(n) \sim C_{k,r} X^{D+1},$$

where

$$C_{k,r} = \frac{H_{k,r}(D+1)}{D+1} > 0, \quad D = r(k-r).$$

Proof. The Dirichlet series (31) has a single rightmost simple pole at $s = D + 1$ with residue $H_{k,r}(D+1)$. Since the coefficients are nonnegative, a standard Tauberian theorem yields the stated asymptotic. $\square$

This gives a geometric rule:

$$\text{cumulative exponent} = 1 + \dim \operatorname{Gr}(r, k) = 1 + r(k-r).$$

The same construction can be repeated with flag varieties or other smooth homogeneous spaces over the rings $\mathbb{Z}/p^e\mathbb{Z}$ whenever their lifting counts are uniform.

10 Pratt orbit systems

We now replace the divisor order by a genuinely different incidence geometry. Let $m_q(n)$ denote the complete Pratt coordinate counting occurrences of the prime label q in the unfolded Pratt forest of n . Define

$$a \leq_P b \iff m_q(a) \leq m_q(b) \text{ for every prime } q.$$

The poset $(\mathbb{N}, \leq_P)$ is locally finite because $a \leq_P b$ implies $a \leq b$ in ordinary size. The Pratt-Moebius theory developed for these coordinates gives, for every integer $k \geq 1$, primitive masses $C_k(d)$ satisfying

$$x^k = \sum_{d \leq_P x} C_k(d). \quad (32)$$

Equivalently,

$$C_k(d) = \sum_{a \leq_P d} \mu_P(a, d) a^k.$$

We take (32) as the incidence input for the orbit-system construction.

10.1 Primitive Pratt tuples and their unit orbits

A concrete realization of the primitive mass is

$$E_k^P(d) = \left\{ (a_1, \dots, a_k) \in (\mathbb{Z}/d\mathbb{Z})^k : \text{for every } p \mid d \text{ some } a_i \equiv 0 \pmod{p} \right\}.$$

Then $|E_k^P(d)| = C_k(d)$. Let

$$G_k^P(d) = ((\mathbb{Z}/d\mathbb{Z})^\times)^k$$

act coordinatewise. The condition defining $E_k^P(d)$ is invariant under this action. By theorem 3.5, with

$$X_k^P(x) := \bigsqcup_{d \leq_P x} E_k^P(d),$$

we obtain an orbit system whose total layer size is exactly x^k .

Proposition 10.1 (Pratt orbit count). *If $d = \prod_{p^e \parallel d} p^e$, then*

$$A_k^P(d) := \#(E_k^P(d)/G_k^P(d)) = \prod_{p^e \parallel d} ((e+1)^k - 1).$$

Proof. For one coordinate modulo p^e , multiplication by units preserves and is transitive on the classes with fixed valuation

$$v_p(\gcd(a_i, p^e)) \in \{0, 1, \dots, e\}.$$

Hence a local orbit is determined by a k -tuple of these $e+1$ valuation levels. The primitive Pratt condition excludes exactly the all-zero valuation tuple, leaving $(e+1)^k - 1$ local orbits. The Chinese remainder theorem makes the prime-power components independent, so the orbit counts multiply. $\square$

For example,

$$A_k^P(p) = 2^k - 1, \quad A_k^P(p^2) = 3^k - 1, \quad A_k^P(pq) = (2^k - 1)^2.$$

Thus the primitive mass depends on the actual primes through $C_k(d)$, while the quotient atom count depends only on the exponent pattern of d .

10.2 Worked Pratt examples

Take $k = 2$. For $d = 7$ the primitive set consists of pairs $(a, b) \in \mathbb{F}_7^2$ with at least one zero coordinate. Hence

$$|E_2^P(7)| = 7^2 - 6^2 = 13.$$

Under $(\mathbb{F}_7^\times)^2$ these 13 points form exactly three orbits,

$$(0, 0), \quad (0, *), \quad (*, 0),$$

so $A_2^P(7) = 3$.

For $d = 49$ the primitive mass is

$$|E_2^P(49)| = 7^2(7^2 - 6^2) = 637,$$

whereas the orbit count is only

$$A_2^P(49) = 3^2 - 1 = 8.$$

For $d = 21 = 3 \cdot 7$, Chinese remainder factorization gives

$$|E_2^P(21)| = (3^2 - 2^2)(7^2 - 6^2) = 65, \quad A_2^P(21) = 3 \cdot 3 = 9.$$

These examples show that the Pratt primitive mass and the Pratt orbit count carry different information: the former remembers prime sizes, while the latter remembers the local valuation-pattern geometry.

10.3 A polylogarithmic-linear Pratt PNT

Set $r_k = 2^k - 1$. The atom-count Dirichlet series is

$$Z_k^P(s) = \sum_{n \geq 1} \frac{A_k^P(n)}{n^s} = \prod_p \left(1 + \sum_{e \geq 1} \frac{(e+1)^k - 1}{p^{es}} \right).$$

The local factor begins

$$1 + \frac{r_k}{p^s} + O_k(p^{-2\Re s}).$$

Hence

$$Z_k^P(s) = \zeta(s)^{r_k} H_k^P(s), \tag{33}$$

where H_k^P is represented by an absolutely convergent Euler product in a neighborhood of $s = 1$ and $H_k^P(1) > 0$.

Theorem 10.2 (Pratt orbit prime-number law). *For every fixed $k \geq 1$,*

$$\sum_{n \leq X} A_k^P(n) \sim C_k^P X(\log X)^{2^k - 2},$$

where

$$C_k^P = \frac{H_k^P(1)}{\Gamma(2^k - 1)} > 0.$$

Proof. Equation (33) has the standard Selberg–Delange form $\zeta(s)^{r_k} H(s)$ with nonnegative coefficients and H analytic and nonzero at 1. The Selberg–Delange theorem gives

$$\sum_{n \leq X} A_k^P(n) \sim \frac{H_k^P(1)}{\Gamma(r_k)} X(\log X)^{r_k - 1}.$$

Substitute $r_k = 2^k - 1$. □

Thus the Pratt construction produces a growth regime not present in the previous examples:

$$\boxed{X(\log X)^{2^k - 2}}.$$

For $k = 1$ the law is linear; for $k = 2$ it is $CX(\log X)^2$; for $k = 3$ it is $CX(\log X)^6$.

Remark 10.3 (Support quotient). If the p -adic valuation level is forgotten and only the nonempty set of coordinates divisible by p is retained, then the atom count becomes

$$B_k(n) = (2^k - 1)^{\omega(n)}.$$

Its Euler product has the same leading $\zeta(s)^{2^k-1}$ singularity, so it has the same logarithmic exponent. This suggests that the exponent $2^k - 2$ is controlled by the number of first-level local orbit types rather than by higher p -adic details.

11 Products of concrete orbit systems

The abstract product theorem becomes useful when applied to the new families.

11.1 Grassmann $\times$ Pratt

Take the Grassmann system (k, r) and Pratt system of tuple rank h . In the rectangular gauge $n_1, n_2 \leq X$, the product atom count is

$$A_{\text{Gr} \times P}(X, X) = A_{k,r}^{\text{Gr}}(X) A_h^P(X).$$

Combining theorems 9.2 and 10.2 gives

$$A_{\text{Gr} \times P}(X, X) \sim C X^{r(k-r)+2} (\log X)^{2^h-2}.$$

Thus a Cartesian product combines polynomial geometric dimension with Pratt polylogarithmic complexity.

11.2 Jordan $\times$ Jordan

For Jordan ranks k and ℓ , rectangular counting gives

$$A_{J_k \times J_\ell}(X, X) \sim C_k C_\ell X^{k+\ell}.$$

This is a genuine product law rather than a new Jordan rank: the atoms retain both orbit coordinates.

11.3 Exponential $\times$ exponential

For two necklace systems on alphabets of sizes q_1, q_2 , the bivariate type count is

$$a(n_1, n_2) = M_{q_1}(n_1) M_{q_2}(n_2) \sim \frac{q_1^{n_1} q_2^{n_2}}{n_1 n_2}.$$

On the diagonal $n_1 = n_2 = n$ this becomes

$$a(n, n) \sim \frac{(q_1 q_2)^n}{n^2}.$$

The n^{-2} dilution is therefore generated by a product construction, not inserted by hand.

12 Cyclotomic roots: an exact linear system

Let

$$X_n = \mu_n = \{z \in \mathbb{C} : z^n = 1\}.$$

The primitive type- d layer is the set of primitive d th roots of unity, of cardinality $\varphi(d)$. The Galois group

$$G_d \cong (\mathbb{Z}/d\mathbb{Z})^\times$$

acts freely and transitively on this layer. Thus there is exactly one orbit atom of every type:

$$a(d) = 1.$$

The master identity is the classical formula

$$n = \sum_{d|n} \varphi(d).$$

With gauge $\nu(d) = d$,

$$A(X) = \sum_{d \leq X} a(d) = \lfloor X \rfloor \sim X.$$

This is an exact linear cumulative law. It is also the $k = 1$ numerical shadow of the Jordan system, but the geometric realization is different.

13 Natural numbers: the Pratt measure

The ordinary integers do not supply a nontrivial orbit quotient analogous to Frobenius or word rotation, but their measured structure can be made intrinsic rather than chosen ad hoc. Let $M(n) = (m_q(n))_q$ be the numerical Pratt vector: $m_q(n)$ counts occurrences of the prime label q in the unfolded Pratt forest of n . The Pratt reconstruction identity is

$$n = \prod_q \left(\frac{q}{q-1} \right)^{m_q(n)}.$$

Hence the positive linear functional

$$L_P(x) := \sum_q x_q \log \frac{q}{q-1} \tag{34}$$

satisfies the exact identity

$$L_P(M(n)) = \log n. \tag{35}$$

Thus logarithmic size is the total mass of the Pratt forest when every node labelled q is assigned the local mass

$$\varepsilon_q := \log \frac{q}{q-1} > 0.$$

For a prime p , write T_p for its Pratt tree. Then

$$\text{wt}(T_p) = \sum_q m_q(p) \varepsilon_q = \log p.$$

We now realize the divisor-poset orbit system without changing OS1–OS4. For every prime power p^m , let E_{p^m} be a singleton decorated by a copy of T_p , and give that singleton the intrinsic mass

$$\mu(E_{p^m}) = \text{wt}(T_p) = \log p.$$

For non-prime-powers put $E_d = \emptyset$, and let all groups be trivial. For $n \geq 1$ define

$$X_n := \bigsqcup_{p^m | n} E_{p^m}.$$

Then

$$\mu(X_n) = \sum_{p^m | n} \text{wt}(T_p) = \sum_{p^m | n} \log p = \sum_p v_p(n) \log p = \log n. \quad (36)$$

Equivalently,

$$\log n = \sum_{d|n} \Lambda(d), \quad \Lambda(p^m) = \log p,$$

with $\Lambda(d) = 0$ otherwise. Thus the classical von Mangoldt decomposition is the divisor-poset primitive decomposition of the same logarithmic mass that is reconstructed internally by the Pratt linear functional.

Proposition 13.1 (Pratt origin of the integer measure). *The measured integer orbit system has total-layer function*

$$F(n) = L_P(M(n)) = \log n,$$

and primitive mass

$$P_0(d) = \Lambda(d).$$

In particular, the measure is not selected to force the von Mangoldt identity: it is inherited from the Pratt reconstruction of n .

Proof. Equation (35) gives $F(n) = \log n$. Applying ordinary divisor-poset Moebius inversion to F gives

$$P_0(d) = \sum_{e|d} \mu(e) \log(d/e) = \Lambda(d).$$

The displayed construction of X_n realizes these primitive masses by Pratt-tree components. □

The group action in this example is deliberately trivial: the intrinsic new ingredient is the measure, not a hidden symmetry quotient. There is one atom in every prime-power layer. Thus the unweighted cumulative atom count is

$$Q(X) := \#\{p^m \leq X\} = \pi(X) + O(X^{1/2}).$$

The classical prime number theorem gives the norm-logarithmic law

$$Q(X) \sim \pi(X) \sim \frac{X}{\log X}.$$

The weighted primitive count is the Chebyshev function

$$\psi(X) = \sum_{d \leq X} \Lambda(d),$$

and the same theorem is equivalent to the linear measured law

$$\psi(X) \sim X.$$

Hence the PNT is unchanged; what changes is the geometric source of the measure. The old auxiliary weighting is replaced by the canonical positive Pratt functional (34).

14 Burnside, necklaces, and Witt vectors

Every finite G -set decomposes uniquely into transitive G -orbits. The Burnside ring records such orbit decompositions, while marks or fixed-point counts provide another coordinate system. In the procyclic setting, necklace coordinates and Witt-vector ghost coordinates are closely related. Metropolis–Rota developed the necklace algebra in relation to Witt vectors, and Dress–Siebeneicher interpreted Witt vectors through completed Burnside rings of profinite groups.

This existing theory is an algebraic foundation, not a universal asymptotic theorem. In orbit-system language:

- Burnside/Witt theory supplies a container for converting fixed-point or ghost data into orbit data;
- an orbit system additionally specifies a family of layers and a proper gauge;
- the growth law then depends on those analytic inputs.

For example, procyclic ghost counts q^n recover necklace coordinates $M_q(n)$ and hence the exponential law. A fixed finite G -set has only finitely many orbit atoms and therefore no nontrivial asymptotic law. The algebraic container is common; the growth regime is not.

This distinction is one of the motivations for OS4: asymptotics should not be silently built into the orbit algebra.

15 Comparison: the common geometry

The abstract geometric objects are now visible in every example:

$$\boxed{X_t \supset E_t \longrightarrow E_t/G_t = \mathcal{A}_t \xrightarrow{\nu} \text{growth.}}$$

Here X_t is the total layer, E_t is the exact or primitive stratum, G_t is the symmetry group (trivial in the integer example), $\mathcal{A}_t$ is the orbit-atom space, and ν is the counting gauge.

System	Total layer X_t / $F(t)$	Primitive stratum	Symmetry / atoms	Prime-number law
Finite fields	$\mathbb{F}_{q^n}$, $F(n) = q^n$	exact degree d	Frobenius, orbit size d	$I_q(d) \sim q^d/d$ (exponential layer)
Necklaces	q -ary words, $F(n) = q^n$	least period d	rotation, orbit size d	$M_q(d) \sim q^d/d$ (exponential layer)
Periodic dynamics	$\text{Fix}(T^n)$	least period d	iteration by C_d	$c\lambda^d/d$ under exponential fixed-point growth
Graph shift	based closed walks, $\text{tr}(B^n)$	least-period walks	cyclic rotation	λ^d/d for primitive B
Jordan k	$(\mathbb{Z}/n\mathbb{Z})^k$, $F(n) = n^k$	exact additive order d	scalar units, size $\varphi(d)$	$\sum_{d \leq X} A_k(d) \sim C_k X^k$ (polynomial cumulative)
Grassmann (k, r)	incidence closure of surjections	surjective maps	GL_r , kernels are Grassmann atoms	$CX^{r(k-r)+1}$ (polynomial cumulative)
Pratt k	X_x^P , $F(x) = x^k$ on $(\mathbb{N}, \leq_P)$	Pratt-Moebius layer $C_k(d)$	coordinatewise units	$CX(\log X)^{2^k-2}$ (polylog-linear)
Cyclotomic	μ_n , $F(n) = n$	exact order d	Galois, size $\varphi(d)$	$a(d) = 1$, cumulative $\sim X$
Natural numbers	Pratt-measured X_n , $F(n) = L_P(M(n)) = \log n$	prime powers	trivial orbit; mass $\text{wt}(T_p) = \log p$	$Q(X) \sim X/\log X$, $\psi(X) \sim X$
$\mathbb{Q}[x]$, fixed degree d	Mahler ball $\mathcal{M}(f) \leq T$	irreducibles / Pratt-primitive Mahler mass	polynomial Pratt order	Mahler: $C_d T^d$; Pratt-Mahler: conjecturally $K_d T^d$ for $d \geq 2$

Two lessons follow.

First, equal orbit transforms can come from different geometries. Finite fields and necklaces have identical $(F, w) = (q^n, n)$ and therefore identical atom counts, despite different underlying objects.

Second, different growth laws are controlled by different inputs. Exponential layers with linear orbit weight give λ^n/n ; polynomial layers can give polynomial cumulative laws; logarithmic Pratt-measured layers can have sparse Moebius transforms and require an independent norm theorem.

The new examples add two further lessons. A geometric quotient of dimension D can produce a cumulative law of degree $D + 1$, while a multiplicative local orbit count with r first-order types can produce a pole of order r and hence an $X(\log X)^{r-1}$ law. Cartesian products multiply atom counts and therefore add the corresponding growth exponents.

Thus the useful “knobs” are

$$\boxed{P, \quad F, \quad P_0, \quad G_t\text{-orbit geometry,} \quad \nu, \quad \times.}$$

Here P is now itself variable: changing the incidence poset from divisibility to the Pratt order changes the primitive transform before any orbit quotient is taken.

16 Relation to recursive factorization systems

An orbit system does not require a factorization monoid and does not require a predecessor map. A ranked recursive factorization system asks a different question: given an atom, which smaller atoms occur in its recursive predecessor?

The distinction is therefore simple. An orbit system asks which primitive stratum is present, which symmetry acts on it, and how the quotient atoms grow. A recursive factorization system asks which lower-rank atoms occur in the predecessor of a given atom. Neither structure canonically determines the other. If desired, one may adjoin the free commutative monoid on $\bigsqcup_t \mathcal{A}_t$ and then specify an independent rank-decreasing predecessor D . The trivial choice $D(a) = 1$ is always available but carries no recursive information.

This separation is important: the divisor stratification in an orbit system should not be mistaken for a canonical derivative. Conversely, a Pratt-type predecessor may exist even when no useful primitive-orbit decomposition has been chosen.

17 Conclusion

The orbit-system viewpoint separates four operations:

$$\boxed{\text{measured layer} \longrightarrow \text{primitive stratum} \longrightarrow \text{symmetry quotient} \longrightarrow \text{gauge}.}$$

The expanded definition makes two points explicit. First, the incidence poset is part of the geometry and need not be divisibility. Second, orbit systems admit Cartesian products, so new atom-growth laws can be generated functorially from old ones.

The classical examples remain instructive: finite fields, necklaces, and periodic dynamics explain exponential λ^n/n laws; Jordan systems explain polynomial cumulative laws; the integer example derives $\log n$ from the positive Pratt functional $L_P(M(n))$ and then recovers Λ by divisor-poset Moebius inversion. The new constructions enlarge this picture. Grassmann orbit systems produce

$$X^{1+r(k-r)},$$

with exponent determined by Grassmann dimension. Pratt orbit systems use the Pratt poset and its Moebius inversion and produce

$$X(\log X)^{2^k-2}.$$

Products then combine these regimes, for example

$$X^{D+2}(\log X)^m$$

in a Grassmann–Pratt rectangle, or $(q_1 q_2)^n/n^2$ on the diagonal of two exponential systems.

For monic polynomials over $\mathbb{Q}$, Mahler geometry and Pratt geometry coexist on the same atom universe: the former gives the classical fixed-degree T^d law, while the latter gives the signed Chebyshev transform $\Psi_{P,d}$ and the empirical conjecture $\Psi_{P,d}(T) \sim K_d T^d$ for $d \geq 2$.

The common operation is not multiplication but primitive quotienting. Multiplication re-enters at a second level: as the Cartesian product operation on orbit systems and, in arithmetic examples, through Euler products used to analyze atom counts. This distinction lets incidence geometry, symmetry, and asymptotic growth vary independently.

The next questions are structural. Which locally finite posets admit nonnegative Moebius primitive masses for natural layer functions? Which geometric quotient families have a dimension-controlled PNT? Which product operations preserve stronger error terms or Riemann-hypothesis-type statements? And how do orbit systems interact with a separate ranked recursive predecessor when both structures are available on the same objects?

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